

Question	Marking Guidance	Mark	Comments
04.1	1. (The vesicle) fuses/binds with a lysosome; 2. Lysozyme/s hydrolyses/digests (SCFR) OR Hydrolytic enzyme/s breaks down/hydrolyses/ digests (SCFR);	2 (2 x AO2)	2. Accept protease hydrolyses/digests (SCFR);

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04.2	Control 1. Cardiomyocytes/(cardiac muscle) cells have not been replaced OR Infarcted tissue is not repaired/replaced OR (The contraction of the ventricle is weak as) there is a small number of cardiomyocytes/(cardiac muscle) cells still alive OR (The ventricular pressure is low as) cardiomyocytes/(cardiac muscle) cells are damaged/dead; 2. (Pressure is not zero as) not all cardiomyocytes/(cardiac muscle) cells died OR (Pressure is not zero as) not all cardiomyocytes/(cardiac muscle) cells became infarcted tissue; c-KIT- 3. Higher than control, so (some) stem cells (must) have been able to differentiate OR Higher than control, so (some) cardiomyocytes/(cardiac muscle) cells have been replaced/infarcted tissues have been repaired; 4. (So) <i>c-KIT</i> is not the only gene responsible for differentiation OR (So) SCF must be able to bind to something other than SCFR OR (So) something else must be able to activate TK in cells; 5. Increase is less than c-KIT+ group as they could not make SCFR OR Increase is less than c-KIT+ group as they could not activate TK;	4 (4 x AO3)	1. Reject 'some stem cells from bone marrow moved to the infarcted tissue'

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04.3	A correct answer 34% = 2 marks;; Evidence of 2 and 1 = 1 mark An answer of 42.8/43% = 1 mark (answer that did not subtract the 'control' from read values)	2 (2 x AO2)	

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04.4	Connexin-43 1. (Connexin-43) allows impulses to pass to the bottom/apex of the heart/ventricles OR (Connexin-43) allows impulses to pass through Purkyne tissue/the bundle of His; GATA-4 2. More actinomyosin bridges OR More binding sites on actin OR More myosin filaments/heads;	2 (2 x AO3)	1. Accept (connexin-43) allows diffusion/movement of ions (between cardiomyocytes) 1. Ignore references to signals/information/messages 2. Accept cross bridges for actinomyosin bridges 2. Accept more binding of myosin to actin
